infosource

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1. Basic Information

Item	Content	Note
Remote repository	https://github.com/MacWaveOrg/infosource	Protected repository
Owner	Sha0huaZhang, https://github.com/Sha0huaZhang	
Old branch	https://github.com/MacWaveOrg/MacWave/tree/legacy-infosource	Archived and abandoned; renamed legacy-infosource — do not push to the old branch
Contact	hi@macwave.org (email); https://github.com/Sha0huaZhang (GitHub)	

To contribute, clone this repository first:

```
mkdir -p ~/src/MacWaveOrg
cd ~/src/MacWaveOrg
git clone https://github.com/MacWaveOrg/infosource.git
```

2. Index Format

A. Package index

Here is a real snippet:

(arm64 is the same, omitted)

Run the following (install the tree package first, e.g. `wave install tree` ; `brew install tree`):

```
tree ~/src/MacWaveOrg/infosource
```

Output (only some packages and dependencies are listed here):

```
~/src/MacWaveOrg/infosource
├── deps
│   ├── depsinfo_amd64
│   │   ├── bzip2
│   │   │   ├── _bzip2@1.0.8
│   │   │   └── _bzip2@common
│   │   ├── _ncurses@6.4
│   │   ├── _ncurses@6.6
│   │   └── _ncurses@common
```

```
│   │   ├── oniguruma
│   │   │   ├── _oniguruma@6.9.9
│   │   │   └── _oniguruma@common
│   │   ├── openssl
│   │   │   ├── _openssl@3.0.15
│   │   │   └── _openssl@common
│   │   ├── icu
│   │   │   ├── _icu@68.1
│   │   │   └── _icu@common
│   │   ├── pcre2
│   │   │   ├── _pcre2@10.42
│   │   │   └── _pcre2@common
│   │   ├── xz
│   │   │   ├── _xz@5.6.4
│   │   │   └── _xz@common
(...)
├── pkg
│   ├── pkginfo_amd64
│   │   ├── bat
│   │   │   ├── _bat@0.25.0
│   │   │   ├── _bat@0.26.0
│   │   │   ├── _bat@0.26.1
│   │   │   └── _bat@common
│   │   ├── btop
│   │   │   ├── _btop@1.4.5
│   │   │   ├── _btop@1.4.6
│   │   │   ├── _btop@1.4.7
│   │   │   └── _btop@common
│   │   ├── choma
│   │   │   ├── _choma@1.0-macwaveteam2
│   │   │   └── _choma@common
│   │   ├── jq
│   │   │   ├── _jq@1.8.2
│   │   │   └── _jq@common
│   │   │   │   ├── trollrestore
│   │   │   ├── _trollrestore@1.0
│   │   │   └── _trollrestore@common
│   │   └── wget
│   │       ├── _wget@1.25.0
│   │       └── _wget@common
│   │
(omitted)
```

Run the following commands:

```
cd ~/src/MacWaveOrg/infosource/pkg/pkginfo_amd64/wget
cat _wget@1.25.0
```

Output:

```
sha256: "11c6d0041823aef0b8b764e6820e846d6bb63b4202e24ca4dd94d343d9f271f9"
url: "https://repo.continuum.io/pkgs/main/osx-64/wget-1.25.0-h3a17b82_0.tar.bz2"
deps: "gettext@0.21.0"
      "libiconv@1.16"
      "libidn2@2.3.4"
      "openssl@3.0.15"
      "pcre2@10.42"
      "zlib@1.2.13"
```

Run the following commands:

```
cd ~/src/MacWaveOrg/infosource/pkg/pkginfo_amd64/bat
cat _bat@0.26.1
```

Output:

```
sha256: "e30beff26779c9bf60bb541e1d79046250cb74378f2757f8eb250afddb19e114"
url: "https://github.com/sharkdp/bat/releases/download/v0.26.1/bat-v0.26.1-aarch64-apple-darwin.tar.gz"
```

Comparing the two, the `deps` field is not mandatory. A package gets its dependency information from the `deps` field and downloads them the same way it downloads a package (only into a different directory). When a package needs dependencies the `deps` field exists; when a package needs no dependencies the `deps` field must not exist.

`_dependency@version` mainly stores information that changes easily, such as SHA256, URL and dependencies. Each field means the following:

Field	Meaning	Purpose	Note
sha256	The SHA256 value of the package	Compared with the package after download to prevent tampering	Must not be empty or wrong, otherwise the installation is aborted and a warning is raised
url	The URL of the package	Used to fetch the package	Must not be empty

Field	Meaning	Purpose	Note
deps	The dependencies of the package	Yields the names and versions of the dependencies the package needs; they are looked up under the corresponding part of `deps/` and downloaded, see the relevant section below	Must not be empty when there are dependencies, and must be empty when there are none

Run the following commands:

```
cd ~/src/MacWaveOrg/infosource/pkg/pkginfo_amd64/wget
cat _wget@common
```

Output:

```
bin_name: "wget"
des: "GNU Wget"
hom: "https://www.gnu.org/software/wget/"
lic: "GPL-3.0"
aut: "GNU Project"
```

`_package@common` mainly stores information that rarely changes. Each field means the following:

Field	Full name	Meaning	Note
bin_name	Binary Name	The name of the executable binary	Must not be empty
des	Description	The description of the package	
hom	Homepage	The homepage of the package	
lic	Licence	The licence of the package; common ones are MIT, GPL and so on	
aut	Author	The author of the package, either a person or an organisation	

B. Dependency index

Run the following commands:

```
cd ~/src/MacWaveOrg/infosource/deps/depsinfo_amd64/icu
cat _icu@73.1
```

Output:

```
    sha256: "fd87b109deef638ca91abf3972516ed588f6d63a32b809943018dd206d3cf2f1"
url: "https://repo.continuum.io/pkgs/main/osx-64/icu-73.1-hcec6c5f_0.tar.bz2"
deps: "libcxx@17.0.6"
des: "ICU - International Components for Unicode"
hom: "https://icu.unicode.org/"
lic: "Unicode-3.0"
aut: "Unicode Consortium"
dep_name: "icu"
```

Run the following commands:

```
    cd ~/src/MacWaveOrg/infosource/deps/depsinfo_amd64/bzip2
cat _bzip2@1.0.8
```

Output:

```
    sha256: "438c40aa11178fe552d6c897b1df6dcee2ba366d27d45756c7fd7ece9c047a1c"
url: "https://repo.continuum.io/pkgs/main/osx-64/bzip2-1.0.8-h6c40b1e_6.tar.bz2"
des: "bzip2 - block-sorting compression library (libbz2)"
hom: "https://sourceware.org/bzip2/"
lic: "BSD-4-Clause"
aut: "Julian Seward"
dep_name: "bzip2"
```

This is broadly similar to a package, and comparing them shows that the `deps` field is not mandatory. As we saw earlier, a package gets its dependency information from `deps` and downloads them the same way it downloads a package (only into a different directory). When a dependency still needs other dependencies the `deps` field exists; when it needs none there is no `deps` field. The program handles this recursively.

`_dependency@version` mainly stores information that changes easily, such as SHA256, URL and dependencies (if any). Each field means the following:

Field	Meaning	Purpose	Note
sha256	The SHA256 value of the dependency	Compared with the dependency after download to prevent tampering	Must not be empty or wrong, otherwise the installation is aborted and a warning is raised
url	The URL of the dependency	Used to fetch the dependency	Must not be empty
deps	The dependencies of the dependency	Yields the names and versions of the dependencies the package needs; they are looked up under the corresponding part of `deps/` and downloaded, see the relevant section below	Must not be empty when there are dependencies, and must be empty when there are none

3. Notes

After modifying or adding an index, please confirm:

1. `sha256` and `url` point to the same version, and the hash is accurate. Prefer URLs from the upstream official channel; avoid mirrors and third-party download addresses of questionable trustworthiness. Prefer the sha256 provided by the official channel; if it does not exist, download the file completely to your machine and compute the sha256 from the original format provided by the site (do not unpack an archive if one is provided).
2. When there are dependencies the `deps` field must not be empty; when there are none the `deps` field must be empty
3. Some included packages are not the official build but a widely used build, such as ldid v2.1.5-procursus7; or, when the official project provides no pre-built package, a build made by the author or by other team members, such as ChOma v1.0-macwaveteam2

4. Contributing

Please submit changes through a Pull Request. For bug reports, field disputes or new architecture requirements, file an Issue or contact the maintainer (hi@macwave.org). Do not keep pushing to the old branch.